

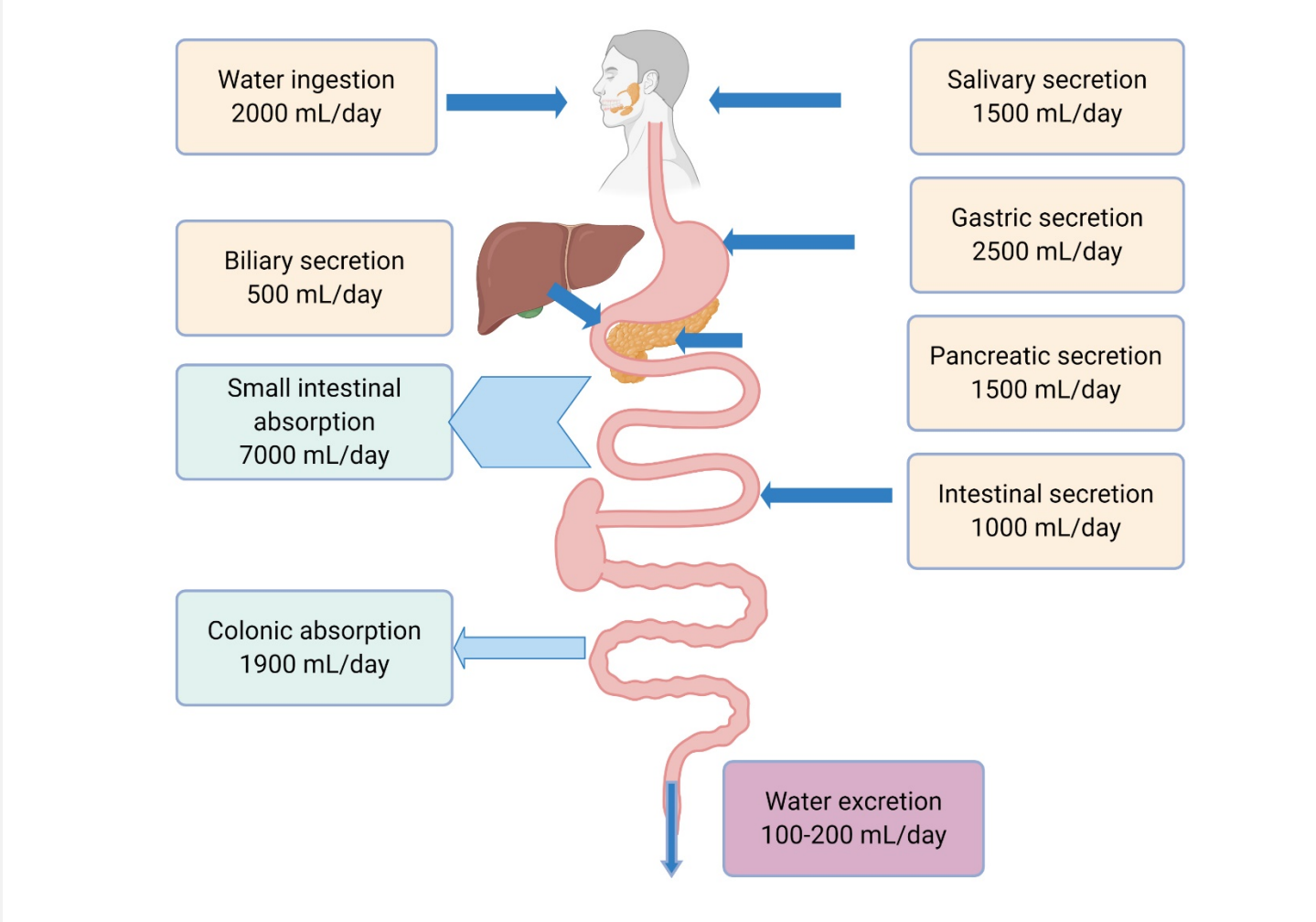
DIARRHEA

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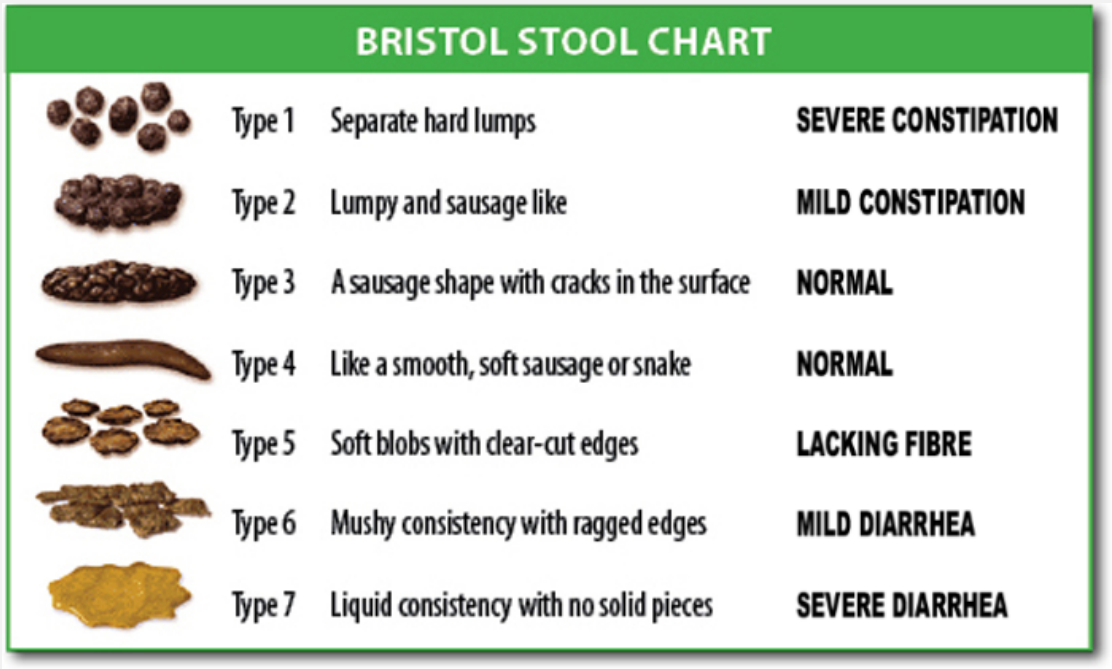
OVERALL FLUID BALANCE IN GI TRACT

Only 100 mL of fluid excreted as feces each day



DIARRHEA

- Definition: ≥ 3 loose BMs daily or total stool weight ≥ 200 g/day
- Duration: ≥ 4 weeks → chronic diarrhea



APPROACH DIARRHEA

- Acute (infectious) vs. chronic (non-infectious)
- Large (right colon/SB) vs. small volume (left colon)
- Watery vs. fatty vs. inflammatory
- Osmotic vs. secretory
- Epidemiologic features: travel hx, PMH (DM, HIV, IBD), recent admission

TYPES OF DIARRHEA

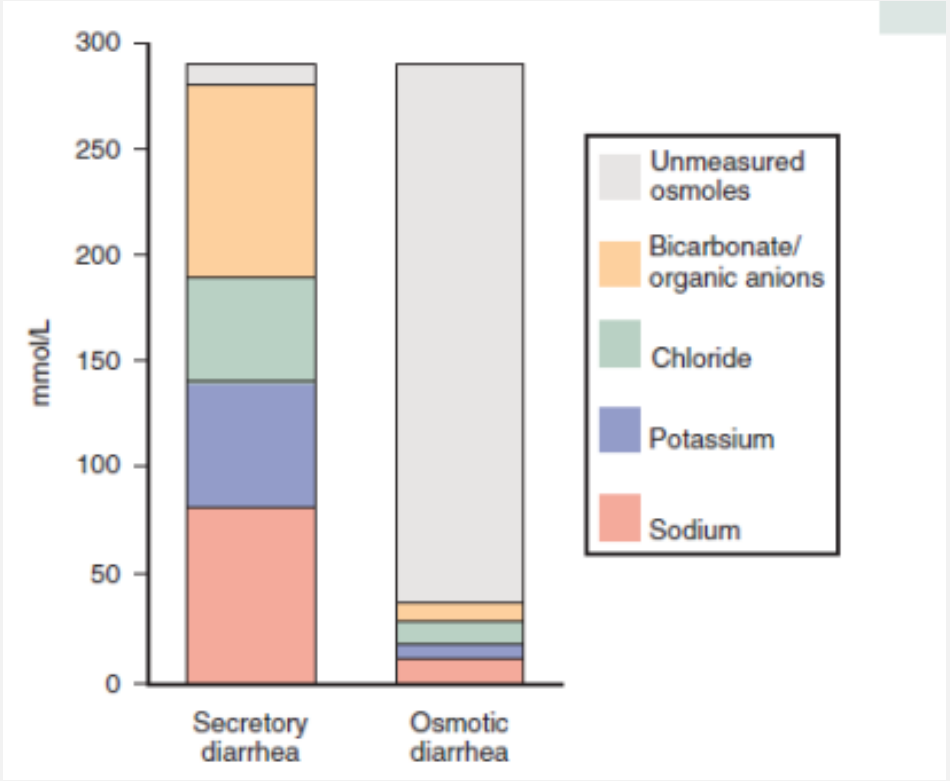
- **Secretory watery diarrhea**: disordered electrolyte transport → ↓ absorption
- **Osmotic watery diarrhea**: poorly absorbed osmotically active substance (e.g., magnesium ion, lactulose) → retains fluid within the lumen to maintain osmotic equilibration with body fluids → ↓ water absorption
- **Steatorrhea** (fatty stool)
- **Inflammatory diarrhea**
- **Dysmotility diarrhea**

SECRETORY VS. OSMOTIC

Type of diarrhea	Causes	Examples
Secretory	Exogenous secretagogues Endogenous secretagogues Loss of intestinal surface area Intestinal ischemia	Enterotoxin (i.e. cholera, E.coli) Neuroendocrine tumors Intestinal resection, celiac disease Diffuse mesenteric atherosclerosis
Osmotic	Ingestion of poorly absorbed agent Reduced nutrient transport	Magnesium ingestion Lactase deficiency

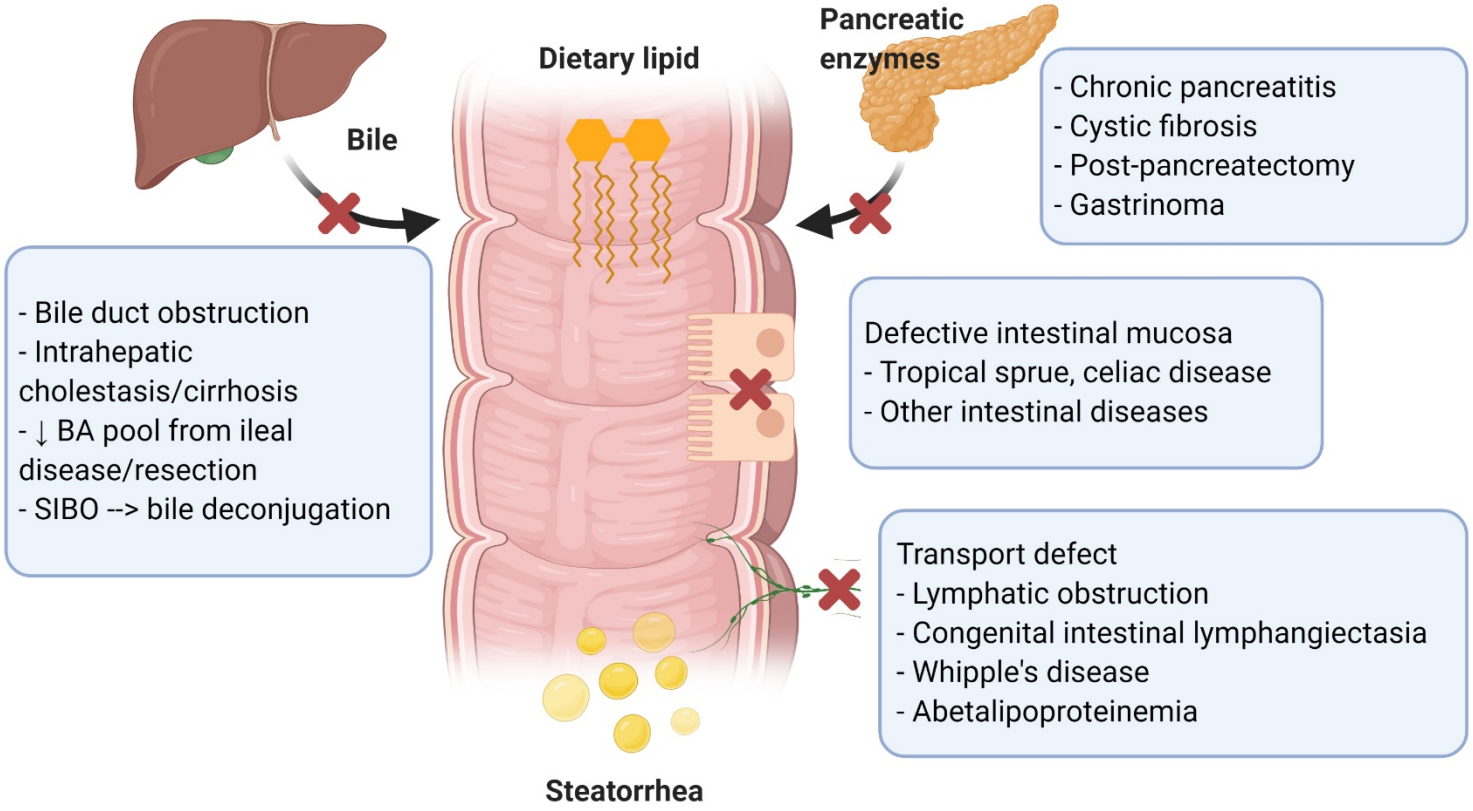
STOOL OSMOTIC GAP

- = 290 – 2x [Na + K]
- Small osmotic gap (<50 mOsm/kg) signifies that the osmolality of stool water is due mostly to incompletely absorbed electrolytes → secretory diarrhea
- Large gap (>100 mOsm/kg) → osmotic diarrhea
- Negative osmotic gap → an excess of cations obligated by ingestion of poorly absorbed multivalent anions (phosphate, sulfate)



STEATORRHEA

Causes



INFLAMMATORY DIARRHEA

Causes

- Inflammatory bowel disease (Crohn's, chronic ulcerative colitis)
- Lymphocytic and collagenous colitis
- Immune-related mucosal disease (1° or 2° immunodeficiencies, food allergy, eosinophilic gastroenteritis, graft-versus-host disease)
- Infections (invasive bacteria, viruses, and parasites)
- Radiation colitis
- Gastrointestinal malignancies: colon cancer, lymphoma

DYSMOTILITY DIARRHEA

Causes

- Irritable bowel syndrome (including post-infectious IBS)
- Visceral neuromyopathies (i.e. diabetic autonomic neuropathy)
- Hyperthyroidism
- Drugs (prokinetic agents)
- Postvagotomy

THANK YOU

